

# Introduction to Causal Inference for Software Engineering

*A 90-minute hands-on tutorial*

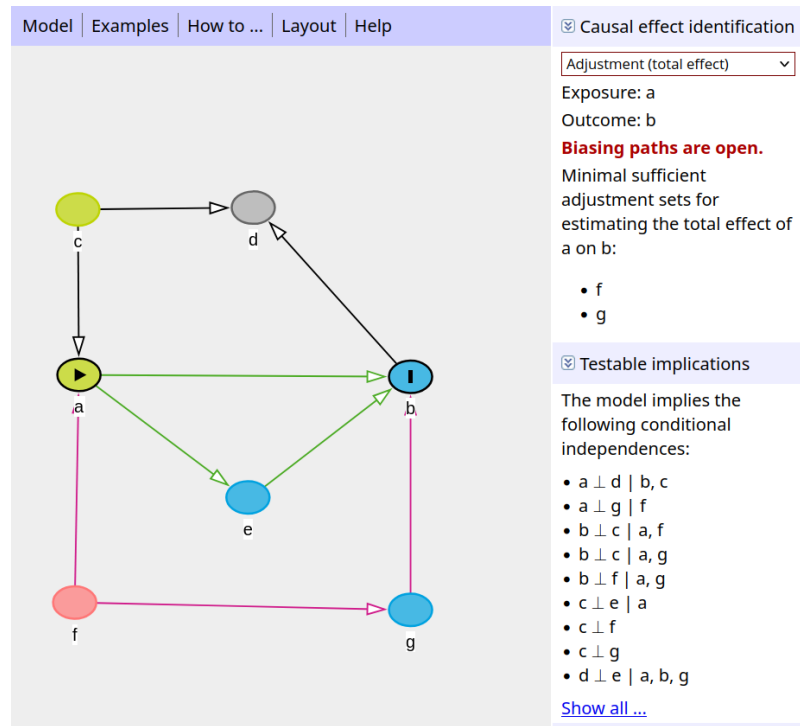
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Section 4

# Exercises: Identification, adjustment sets and conditional independence

# A guides tour through DAGitty

- DAGitty is a browser-based tool to create and analyse DAGs
- It helps causal inference by automatically identifying necessary adjustment sets.
- Based on the graphical causal model, it provides ***conditional independence criteria*** that can be used to test models.
- Access it here: <https://dagitty.net/>



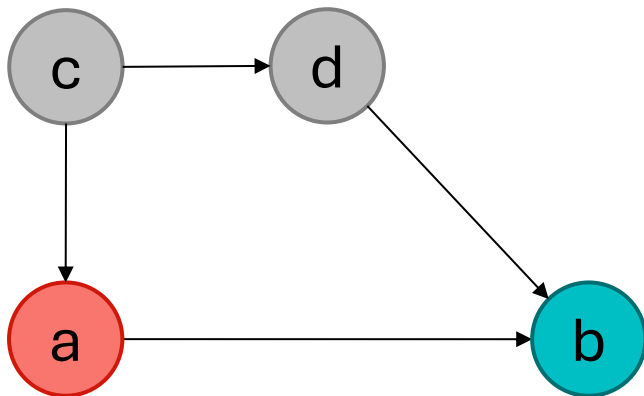
The screenshot shows the DAGitty web application interface. At the top, there is a navigation bar with links: Model, Examples, How to ..., Layout, and Help. The main area displays a directed acyclic graph (DAG) with nodes labeled a through g. Node 'a' is highlighted with a yellow circle and a play button icon. Node 'b' is highlighted with a blue circle and an 'I' icon. The graph shows causal relationships: 'c' points to 'd' and 'a'; 'a' points to 'e'; 'e' points to 'b'; 'f' points to 'g'; and 'g' points to 'b'. The right sidebar contains two sections: 'Causal effect identification' and 'Testable implications'. In the 'Causal effect identification' section, 'Adjustment (total effect)' is selected in a dropdown menu. Below this, it specifies 'Exposure: a' and 'Outcome: b'. A red warning message states 'Biasing paths are open.' followed by 'Minimal sufficient adjustment sets for estimating the total effect of a on b:' and a list of nodes 'f' and 'g'. The 'Testable implications' section states 'The model implies the following conditional independences:' followed by a list of conditional independence statements:  $a \perp d \mid b, c$ ,  $a \perp g \mid f$ ,  $b \perp c \mid a, f$ ,  $b \perp e \mid a, g$ ,  $b \perp f \mid a, g$ ,  $c \perp e \mid a$ ,  $c \perp f$ ,  $c \perp g$ , and  $d \perp e \mid a, b, g$ . A 'Show all ...' link is at the bottom of the sidebar.

Textor et al., [Robust causal inference using directed acyclic graphs: the R package 'dagitty'](#). *International Journal of Epidemiology* 45(6):1887-1894, 2016.

# Live Demo

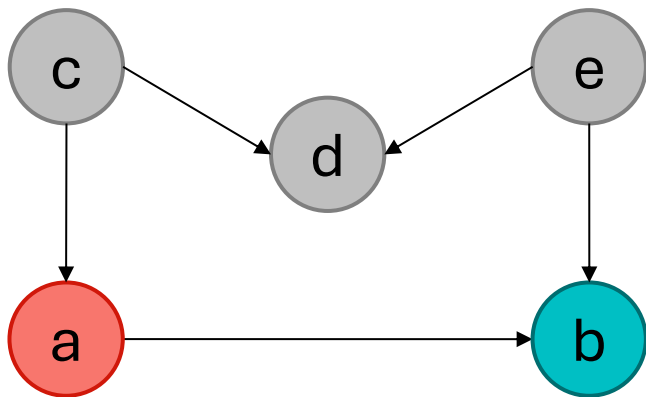
# Problem 1

- Determine the **adjustment set** of variables to block any backdoor causal path for the estimation of the direct effect of a on b.



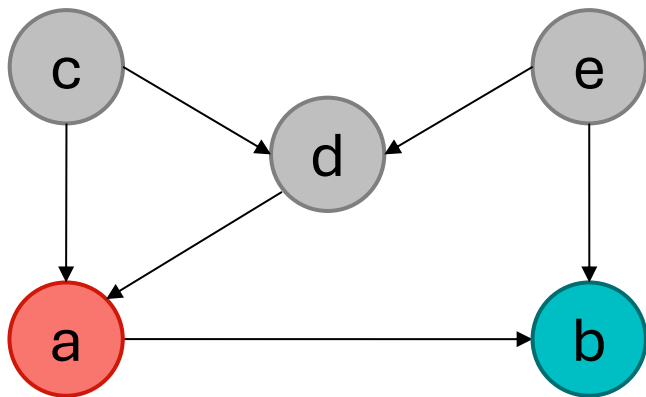
## Problem 2

- Determine the **adjustment set** of variables to block any backdoor causal path for the estimation of the direct effect of a on b.



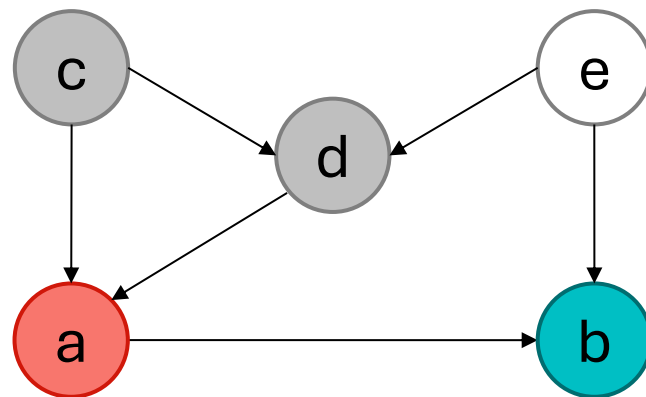
## Problem 3

- Determine the **adjustment set** of variables to block any backdoor causal path for the estimation of the direct effect of a on b.



## Problem 3 more interesting

- Determine the **adjustment set** of variables to block any backdoor causal path for the estimation of the direct effect of a on b.





## Problem 4: Does an automated testing framework (X) reduce post-release defects (Y)?

A software quality team studies the effect of the introduction of an automated testing framework on post-release defects of a large system. They know:

1. Experienced developers are more likely to adopt the framework and to write less defect-prone code.
2. The framework lowers defects directly, and also by raising test coverage.
3. More complex modules tend to have more post-release defects.
4. Both the framework and module complexity raise the number of issues logged.

Draw a DAG including the variables: Autom. Testing, Post-Release Defects, Developer Experience, Test Coverage, Logged Issues, Code Complexity.

Identify:

- Which variables are confounders;
- Which are mediators;
- What is the minimal adjustment set to estimate **the direct and total effect** of automated testing on post-release defects.